

Module 12: Counting Algorithms and Clock Replacement

1. Second Chance (Clock) Replacement

Circular queue with a clock pointer. If Reference bit $R=0 \rightarrow$ Replace. If $R=1 \rightarrow$ Clear $R=0$, give second chance, advance pointer.

2. Enhanced Second Chance 4-Class Model

Class <R, M>	Meaning	Action
(0, 0)	Not recently used, Clean (not modified)	Best candidate for replacement
(0, 1)	Not recently used, Dirty (modified)	Requires disk write before page-out
(1, 0)	Recently used, Clean	Likely in active working set
(1, 1)	Recently used, Dirty	Worst candidate

Second Chance (Clock) Replacement

Clock Pointer inspects Frame

- If $R == 0 \rightarrow$ Replace this page (Victim found!)
- If $R == 1 \rightarrow$ Set $R = 0$, advance pointer (Second Chance!)

Enhanced Second Chance & Counting Algos

Enhanced 4-Class Classification <Reference, Modify>:

- Class 1 (0, 0):** Not used, Not dirty $\rightarrow$ BEST VICTIM!
- Class 2 (0, 1):** Not used, Dirty $\rightarrow$ Needs disk write
- Class 3 (1, 0):** Recently used, Clean $\rightarrow$ Will be used soon
- Class 4 (1, 1):** Recently used, Dirty $\rightarrow$ WORST CANDIDATE

Counting-Based Algorithms:

- LFU (Least Frequently Used):**
Page with smallest reference count is replaced.
- MFU (Most Frequently Used):**
Page with highest count is replaced (newly brought page has count 0).

3. LFU vs MFU

- **LFU:** Smallest reference count replaced.
- **MFU:** Highest reference count replaced (protects new arrivals).